

IGNOU BCS-011 Solved Question Paper

Computer Basics and PC Software

Term-End Examination: December 2021

Time: 3 Hours

Maximum Marks: 100

Weightage: 75%

Question No. 1 is compulsory and carries 40 marks. Attempt any three questions from the rest.

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Solved Paper

Question 1 *Compulsory question*

(a) Convert the following numbers as asked.

Answer.

(i) **Convert $(756)_{10}$ to binary.**

$$756 = 512 + 128 + 64 + 32 + 16 + 4$$

The powers used are $2^9, 2^7, 2^6, 2^5, 2^4, 2^2$. Hence

$$(756)_{10} = (1011110100)_2.$$

(ii) **Convert $(1.0125)_{10}$ to binary.** The integer part is 1. For the fractional part:

Step	Multiply by 2	Bit
1	$0.0125 \times 2 = 0.025$	0
2	$0.025 \times 2 = 0.05$	0
3	$0.05 \times 2 = 0.10$	0
4	$0.10 \times 2 = 0.20$	0
5	$0.20 \times 2 = 0.40$	0
6	$0.40 \times 2 = 0.80$	0
7	$0.80 \times 2 = 1.60$	1
8	$0.60 \times 2 = 1.20$	1
9	$0.20 \times 2 = 0.40$	0

After this, $0.20 \rightarrow 0.40 \rightarrow 0.80 \rightarrow 0.60 \rightarrow 0.20$ repeats. Therefore

$$(1.0125)_{10} = (1.0000001100\dots)_2 = (1.000000(1100))_2.$$

(iii) **Convert $(10110111)_2$ to decimal.**

$$(10110111)_2 = 1 \cdot 128 + 0 \cdot 64 + 1 \cdot 32 + 1 \cdot 16 + 0 \cdot 8 + 1 \cdot 4 + 1 \cdot 2 + 1 \cdot 1 = 183.$$

Thus $(10110111)_2 = (183)_{10}$.

(iv) **Convert $(101111111100111)_2$ to hexadecimal.** Group four bits from the right:

$$1011 \ 1111 \ 1110 \ 0111$$

$$1011 = B, \quad 1111 = F, \quad 1110 = E, \quad 0111 = 7.$$

Hence

$$(101111111100111)_2 = (BFE7)_{16}.$$

(v) **Convert $(49AF)_{16}$ to binary.** Convert each hexadecimal digit into 4 bits:

$$4 = 0100, \quad 9 = 1001, \quad A = 1010, \quad F = 1111.$$

Therefore

$$(49AF)_{16} = (0100100110101111)_2.$$

(b) Explain how disk storage capacity and density are calculated. Also calculate the storage capacity and storage density of the given disk.

Answer. **Storage capacity** of a disk is the total amount of data that can be stored on all recording surfaces. The general formula is

$$\text{Capacity} = \text{surfaces} \times \text{tracks per surface} \times \text{sectors per track} \times \text{bytes per sector}.$$

Given:

$$\text{surfaces} = 10, \quad \text{tracks/surface} = 5000, \quad \text{sectors/track} = 128, \quad \text{sector size} = 1 \text{ kB}.$$

Therefore

$$\text{Capacity} = 10 \times 5000 \times 128 \times 1 \text{ kB} = 6,400,000 \text{ kB}.$$

This is approximately

$$6,400,000 \text{ kB} \approx 6.4 \text{ GB}$$

if $1 \text{ kB} = 1000 \text{ bytes}$. Using $1 \text{ kB} = 1024 \text{ bytes}$, the value is about 6.10 GiB .

Storage density means amount of storage per unit surface area. For one circular recording surface of diameter 3.0 inches, radius is 1.5 inches. Area of one surface is

$$A = \pi r^2 = \pi (1.5)^2 = 2.25\pi \approx 7.07 \text{ in}^2.$$

For 10 recording surfaces,

$$A_{\text{total}} \approx 70.7 \text{ in}^2.$$

Thus approximate storage density is

$$\frac{6,400,000 \text{ kB}}{70.7 \text{ in}^2} \approx 90,500 \text{ kB/in}^2 \approx 90.5 \text{ MB/in}^2.$$

(c) Compare and contrast the following.

Answer.

(i) **Impact printers and non-impact printers**

Impact printer	Non-impact printer
Prints by physically striking an inked ribbon against paper.	Prints without striking the paper.
Noisier because of mechanical impact.	Quieter in operation.
Usually slower and lower quality.	Usually faster and higher quality.
Can produce carbon copies or multipart forms.	Normally cannot produce carbon copies by impact.
Examples: dot matrix, daisy wheel.	Examples: inkjet, laser, thermal printer.

(ii) **Analog monitor and digital monitor**

Analog monitor	Digital monitor
Receives continuously varying analog signals.	Receives discrete digital signals.
Older CRT displays commonly used analog video signals.	Modern LCD/LED monitors commonly use digital interfaces such as DVI, HDMI or DisplayPort.
Signal may degrade due to noise.	More accurate display because digital signals are less affected by noise.
Requires conversion if source is digital.	Better suited to modern digital computers.

(d) You bought a raw computer with only hardware and device drivers. Which essential software is needed to make it usable? Explain any four functions.

Answer. The essential software is an **Operating System (OS)** such as Windows, Linux or macOS. Device drivers alone allow hardware devices to communicate, but the computer becomes usable only when an operating system is installed.

Important functions of an operating system are:

1. **Process management:** It loads programs, schedules CPU time, starts and stops processes, and supports multitasking.
2. **Memory management:** It allocates RAM to running programs, frees memory after use, and manages virtual memory.
3. **File management:** It creates, stores, deletes, copies and organizes files and folders on storage devices.
4. **Device management:** It controls input/output devices through drivers and provides a common interface for printers, keyboard, mouse, display, disk and network devices.
5. **User interface and security:** It provides GUI/command interface, login, user accounts, permissions and protection from unauthorized access.

(e) Processing the given tabular data in a programming language.

Answer. The given data is a rectangular table with 3 rows and 3 columns:

$$\begin{bmatrix} 25 & 30 & 15 \\ 5 & 7 & 90 \\ 5 & 9 & 20 \end{bmatrix}.$$

- (i) The programming element used to process such a table is a **two-dimensional array**, for example `int A[3][3]` in C/C++.
- (ii) Each element is a whole number, so the suitable data type is **integer**, e.g. `int`.
- (iii) Repetitive processing can be done using **looping statements**. For a table, nested **for** loops are most suitable:

```
for(i=0;i<3;i++) for(j=0;j<3;j++) process A[i][j];
```

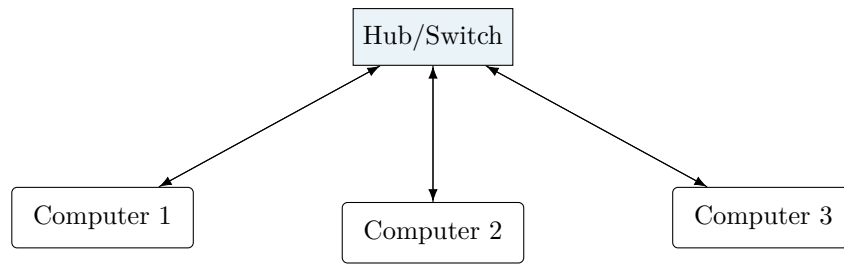
(f) Given $a = 7$ and $b = 9$, find the values.

Answer. Assuming C/C++ style operators:

Expression	Meaning	Value
<code>b % a</code>	Remainder when 9 is divided by 7	2
<code>b++</code>	Post-increment: expression first gives old value, then b becomes 10	expression value 9, later $b = 10$
<code>a > b</code>	Checks whether 7 is greater than 9	false, i.e. 0
<code>a < b b < a</code>	Logical OR of $7 < 9$ and $9 < 7$	true, i.e. 1

(g) Identify the network topology shown in the figure. Explain its advantages and disadvantages.

Answer. The figure shows a central device connected separately to Computer 1, Computer 2 and Computer 3. This is a **star topology**.



Advantages:

1. Easy to install, manage and expand.
2. Failure of one cable or one computer usually does not stop the whole network.
3. Fault detection is easier because each node has a separate connection.
4. Performance is good when a switch is used as the central device.

Disadvantages:

1. The central hub/switch is a single point of failure.
2. Requires more cable than bus topology.
3. Cost increases because of the central device.
4. Network performance depends on the capacity of the central device.

(h) Identify valid IP addresses and find Net_id using subnet masks.

Answer. An IPv4 address has four octets, and each octet must be from 0 to 255. Net_id is obtained by bitwise AND of the IP address and subnet mask.

IP address	Subnet mask	Valid?	Net_id
10.251.250.256	255.255.255.0	Invalid, because 256 is outside 0-255	Not applicable
172.16.20.5	255.240.0.0	Valid	172.16.0.0
192.168.89.25	255.255.0.0	Valid	192.168.0.0
192.168.75.25	255.255.255.0	Valid	192.168.75.0

For example, with 192.168.75.25 and mask 255.255.255.0, the first three octets remain same and host part becomes 0, so Net_id is 192.168.75.0.

(i) Need of browser software, four browser interactions/buttons, and two popular browsers.

Answer. A **browser** is application software used to access, retrieve, interpret and display web pages from the World Wide Web. It communicates with web servers through protocols such as HTTP/HTTPS and presents HTML, images, scripts, audio and video in readable form.

Four common interactions/buttons in a browser are:

1. **Back** - moves to the previous web page.
2. **Forward** - moves to the next page in browsing history.
3. **Refresh/Reload** - reloads the current page.
4. **Address bar/Search bar** - allows entry of URL or search terms.

Two popular browsers are **Google Chrome** and **Mozilla Firefox**. Other examples are Microsoft Edge, Safari and Opera.

Question 2 Computer instruction, CPU, software, LAN and WAN

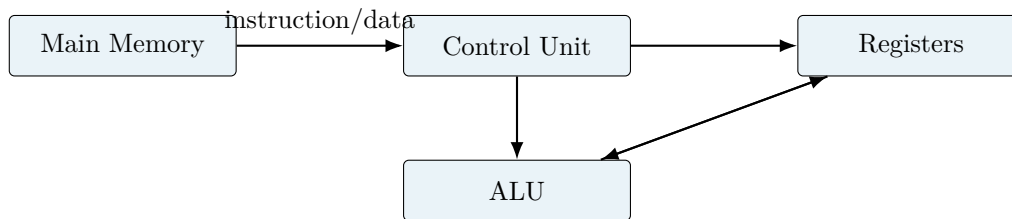
(a) What is an instruction in a computer? Explain with an example. What are the different components of a CPU and their roles? Which component interprets the instruction?

Answer. A **computer instruction** is a binary-coded command that tells the CPU to perform a specific operation. An instruction usually contains an **opcode** and one or more **operands**. The opcode specifies the operation; operands specify the data or memory locations.

Example:

ADD R1, R2

This means add the content of register *R2* to register *R1*. Here, ADD is the opcode and *R1*, *R2* are operands.



Main CPU components are:

1. **Arithmetic Logic Unit (ALU):** Performs arithmetic operations such as addition/subtraction and logical operations such as comparison, AND, OR and NOT.
2. **Control Unit (CU):** Fetches instructions from memory, decodes/interprets them, generates control signals and coordinates all CPU operations.
3. **Registers:** Small, fast storage locations inside the CPU used to hold instructions, addresses, intermediate results and data.
4. **Internal buses:** Carry data, addresses and control signals inside the CPU.

The **Control Unit** interprets or decodes the instruction.

(b) List at least three features of spreadsheet and database software.

Answer.

Spreadsheet software	Database software
Stores data in rows, columns and cells.	Stores data in structured tables, records and fields.
Supports formulas and functions for calculations.	Supports queries for searching and retrieving data.
Can create charts, graphs and pivot summaries.	Provides forms, reports and data validation.
Useful for budgets, marksheets, calculations and analysis.	Useful for large organized data such as student, bank or library records.
Examples: MS Excel, LibreOffice Calc.	Examples: MS Access, MySQL, PostgreSQL, Oracle.

(c) What is LAN? How is it different from WAN? List characteristics, advantages and disadvantages of LAN.

Answer. A **Local Area Network (LAN)** is a computer network covering a small geographical area such as a room, building, office, lab, college department or campus. It allows connected devices to share files, printers, applications and internet access.

LAN versus WAN:

LAN	WAN
Covers small area.	Covers large area such as cities, countries or continents.
Usually owned by one organization.	Often uses telecom/ISP infrastructure.
High speed and low error rate.	Comparatively lower speed and higher latency.
Examples: computer lab network, office network.	Example: Internet, bank network across cities.

Characteristics of LAN:

1. Limited geographical coverage.
2. High data transfer speed.
3. Usually uses Ethernet or Wi-Fi.
4. Devices can share resources through switches, routers and servers.

Advantages: resource sharing, fast communication, centralized backup/security, cost saving through shared printers and internet.

Disadvantages: setup cost, need for administrator, risk of malware spreading quickly, dependence on network devices.

Question 3 *Algorithm, flowchart, e-learning and memory*

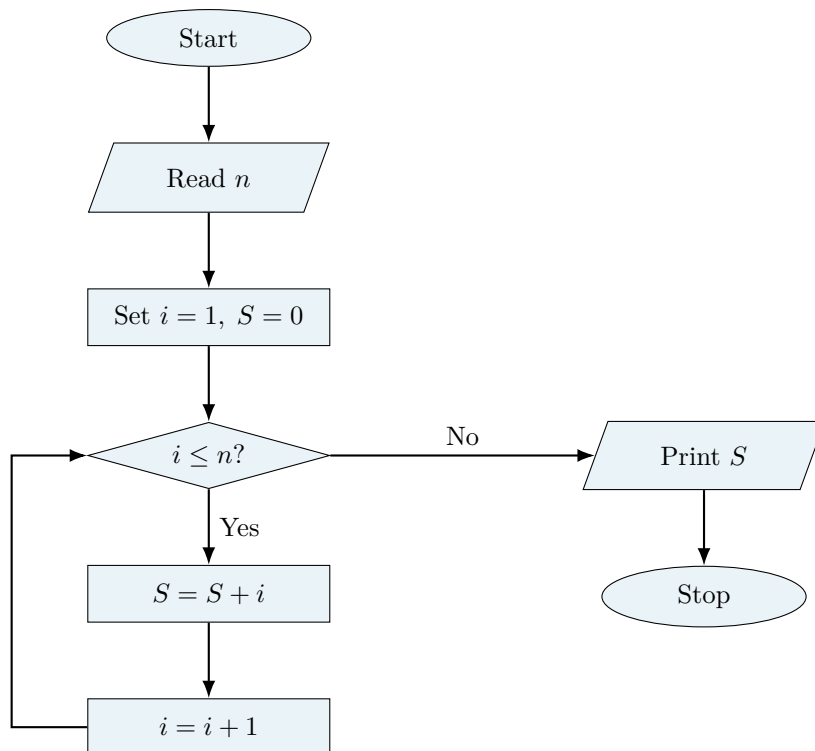
(a) Write an algorithm and draw the flowchart to find the sum of first n natural numbers.

Answer. Algorithm:

1. Start.
2. Read n .
3. Set $i = 1$ and $S = 0$.
4. If $i \leq n$, then set $S = S + i$ and $i = i + 1$; repeat this step.
5. If $i > n$, print S .
6. Stop.

The result can also be checked using the formula:

$$S = 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$



(b) List any six activities that are part of e-learning software. Also list advantages and disadvantages of e-learning.

Answer. Six activities in e-learning software:

1. Online lectures or recorded video lessons.
2. Uploading and downloading study material.
3. Online quizzes and assignments.

4. Discussion forums and chat with teachers/students.
5. Attendance, progress tracking and grading.
6. Online tests, feedback and certificate generation.

Advantages:

1. Learners can study anytime and anywhere.
2. Recorded content can be repeated for revision.
3. Saves travel time and can reduce cost.
4. Learning progress can be tracked digitally.
5. Supports multimedia content such as video, animation and simulations.

Disadvantages:

1. Requires internet, device and basic digital skill.
2. Less face-to-face interaction.
3. Practical/lab-based learning may be difficult.
4. Students may face distraction and lack of discipline.
5. Technical problems can interrupt learning.

(c) What is the purpose of main memory? Explain RAM and ROM. Differentiate between SRAM and DRAM.

Answer. **Main memory** stores the programs and data that the CPU is currently using. It is much faster than secondary storage, so the CPU can quickly fetch instructions and data from it.

RAM (Random Access Memory): RAM is read/write memory used as temporary working memory. It is volatile, meaning its contents are lost when power is switched off. Programs currently running are loaded into RAM.

ROM (Read Only Memory): ROM stores permanent instructions such as firmware or boot program. It is non-volatile, so its contents remain even after power is switched off.

SRAM versus DRAM:

SRAM	DRAM
Static RAM.	Dynamic RAM.
Stores data using flip-flop circuits.	Stores data using capacitors.
Does not need periodic refreshing.	Needs periodic refreshing.
Faster but costlier.	Slower but cheaper.
Used mainly in cache memory.	Used mainly as main memory/RAM.
Consumes more space per bit.	Higher density; stores more bits per chip.

Question 4 *Transmission channels, ports, devices and perverse software*

(a) Compare and contrast the features of twisted pair cable and optical fiber cable; radio waves and microwaves.

Answer.

(i) **Twisted pair cable and optical fiber cable**

Twisted pair cable	Optical fiber cable
Uses copper wires to transmit electrical signals.	Uses glass/plastic fiber to transmit light signals.
Cheaper and easy to install.	Costlier and needs skilled installation.
Lower bandwidth and shorter distance.	Very high bandwidth and long distance.
Affected by electromagnetic interference.	Immune to electromagnetic interference.
Used in telephone lines and Ethernet LANs.	Used in high-speed internet backbone, campus and long-distance links.

(ii) **Radio waves and microwaves**

Radio waves	Microwaves
Lower frequency electromagnetic waves.	Higher frequency electromagnetic waves.
Can travel through buildings and may cover large areas.	Usually require line-of-sight transmission.
Used in radio, TV broadcasting, Wi-Fi and mobile communication.	Used in satellite communication, radar and point-to-point links.
Generally less directional.	Highly directional using dish antennas.

(b) What is a port in the context of a computer? Explain the use of serial port and USB port.

Answer. A **port** is a physical or logical interface through which a computer connects and communicates with external devices or services.

Serial port: A serial port transmits data one bit at a time over a single communication channel. It was commonly used for modems, old mice, serial printers and industrial devices. It is slower than modern USB but useful for simple device communication.

USB port: Universal Serial Bus port is a modern standard port used to connect keyboard, mouse, printer, pen drive, scanner, external hard disk, camera and mobile phone. It supports plug-and-play, faster transfer speeds and can also supply power to devices.

(c) List the characteristics of any two input devices and any two output devices.

Answer.

Type	Device	Characteristics
Input	Keyboard	Enters text, numbers and commands; has alphanumeric, function, navigation and special keys.
Input	Scanner	Converts printed paper or image into digital form; resolution is measured in DPI; useful with OCR.
Output	Monitor	Displays soft copy output; characterized by screen size, resolution, refresh rate and display technology.
Output	Printer	Produces hard copy output; characterized by print speed, resolution, colour support and print technology.

(d) What is perverse software? What is the need of anti-virus software? List two techniques used by anti-virus software to identify viruses. Also list drawbacks.

Answer. **Perverse software** means malicious software that is intentionally designed to harm a computer system, steal data, disturb operation or misuse resources. It includes viruses, worms, Trojan horses, spyware, ransomware and adware.

Need of anti-virus software:

1. Detects and removes viruses and other malware.
2. Scans files, e-mail attachments, downloads and removable media.
3. Protects system files and user data.
4. Provides real-time monitoring and warning against suspicious activity.

Two virus identification techniques:

1. **Signature-based detection:** Compares files with known virus signatures stored in the virus database.
2. **Heuristic/behavior-based detection:** Looks for suspicious behavior or code patterns, even for new or unknown malware.

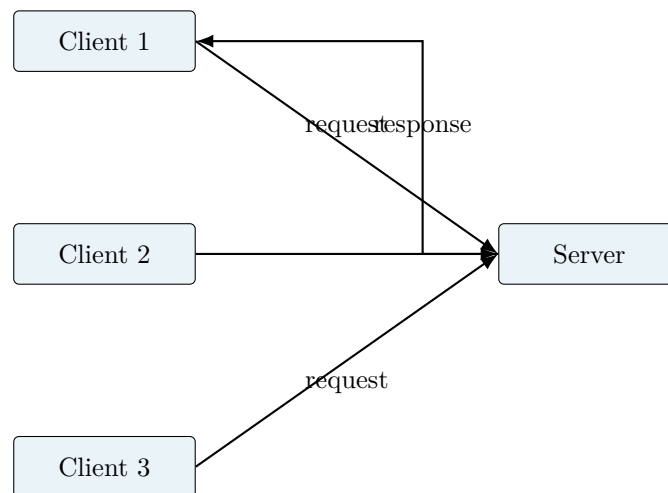
Drawbacks of anti-virus software:

1. It must be updated regularly; otherwise new viruses may not be detected.
2. It may slow down the system during scanning.
3. It can sometimes give false positives or miss new threats.
4. It cannot protect against careless user actions such as clicking unknown links or sharing passwords.

Question 5 *Short notes: explain any five*

(a) Client/server architecture

Answer. Client/server architecture is a network model in which a **client** requests services and a **server** provides those services. For example, a web browser is a client and a web server provides web pages.



Features include centralized service, easier management, better security control and resource sharing. Examples are web servers, mail servers, database servers and file servers.

(b) Open Source Software

Answer. Open source software is software whose source code is available for users to study, modify, improve and redistribute under an open source licence. Examples include Linux, LibreOffice, Mozilla Firefox, Apache web server and Python.

Important features are:

1. Source code is openly available.
2. Users can modify and improve the software.
3. Community participation supports development and bug fixing.
4. It reduces vendor dependence and often reduces cost.
5. Security can improve because code can be reviewed publicly.

(c) Time sharing and multiprogramming systems

Answer. **Multiprogramming** is an operating system technique in which multiple programs are kept in memory at the same time. When one program waits for input/output, the CPU is given to another program. This increases CPU utilization.

Time sharing is an extension of multiprogramming in which many users/programs share the CPU by getting small time slices. The switching is so fast that each user feels that the system is responding interactively.

Multiprogramming	Time sharing
Main aim is better CPU utilization.	Main aim is quick interactive response to users.
Several programs are in memory.	Several users/programs get CPU time slices.
Mostly batch-oriented.	Mostly interactive.
CPU switches mainly when a job waits.	CPU switches after a fixed time quantum.

(d) Collaborations

Answer. Collaboration in web applications means that many users can work together on a shared task using internet-based tools. Examples include Google Docs, Microsoft Teams, shared spreadsheets, GitHub, forums, wikis and learning management systems.

Main features are:

1. Multiple users can create or edit shared content.
2. Version history records changes.
3. Comments and discussion support teamwork.
4. Access permissions control who can view or edit.
5. It supports remote work and real-time communication.

For example, students may prepare a project report together using a shared document, where each student edits a section and the teacher gives comments.

(e) Folders in an e-mail software

Answer. E-mail software organizes messages into folders. Common folders are:

1. **Inbox:** Stores received e-mails.
2. **Sent:** Stores copies of messages sent by the user.
3. **Drafts:** Stores incomplete messages saved for later editing.
4. **Spam/Junk:** Stores suspected unwanted or malicious e-mails.
5. **Trash/Deleted Items:** Stores deleted messages temporarily.

These folders make it easier to manage, search, organize and recover e-mail messages.

(f) ASCII and Unicode

Answer. **ASCII** stands for American Standard Code for Information Interchange. It is a character encoding scheme that represents English letters, digits, punctuation and control characters using 7 bits. For example, capital A has ASCII code 65.

Unicode is a modern character encoding standard designed to represent characters from almost all writing systems of the world, including English, Hindi, Bengali, symbols and emojis. UTF-8 is a common Unicode encoding used on the web.

ASCII	Unicode
Mainly supports English characters.	Supports characters of many languages.
Uses 7 bits in standard form.	Uses variable length encodings such as UTF-8.
Limited character set.	Very large character set.
Example: A = 65.	Example: Hindi characters can be represented.

(g) Disk checker software

Answer. Disk checker software is a utility program used to check storage devices for file system errors, bad sectors and other disk problems. It helps maintain disk health and prevent data loss.

Uses include:

1. Detecting and repairing file system errors.
2. Identifying bad sectors on a disk.

3. Recovering lost clusters or corrupted directory entries.
4. Improving reliability of storage devices.
5. Warning the user about possible disk failure.

Examples include Windows **chkdsk**, ScanDisk and disk checking utilities in Linux.